

LAB # 04a

Manipulation of type casting in Java

Objective

To get familiarity with the type casting in java.

Lab Goals

To provide working insight into the usage of type casting in java programming:

- Demonstrate the implementation of type casting.
- Explore implicit and explicit type casting in Java.

Introduction to Java Type Casting

Type casting is when you assign a value of one primitive data type to another type. In Java, there are two types of casting:

- Widening Casting/ Implicit casting (automatically): Converting a smaller type to a larger type size.

byte -> short -> char -> int -> long -> float -> double

- Narrowing Casting/ Explicit casting (manually): Converting a larger type to a smaller size type.

double -> float -> long -> int -> char -> short -> byte

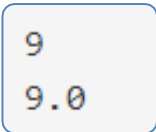
Widening Casting/ Implicit casting

Widening casting is done automatically when passing a smaller size type to a larger size type:

Example#1

```
public class Main {  
    public static void main(String[] args){  
        int myInt =9;  
        double myDouble = myInt; // Automatic casting: int to double  
        System.out.println(myInt); // Output 9  
        System.out.println(myDouble); // Output 9.0  
    }  
}
```

Output



```
9  
9.0
```

Example#2

```
public class Main {  
    public static void main(String[] args) {  
        int numInt = 10;  
        double numDouble = 5.5;  
  
        double result = numInt * numDouble; // Implicit casting from int to double  
        System.out.println("Result: " + result);  
    }  
}
```

Output

```
Result: 55.0
```

Example#3

```
public class Main {  
    public static void main(String[] args) {  
        int numInt = 10;  
        double numDouble = 5.5;  
  
        boolean result = (numInt > numDouble); // Implicit casting from double to int  
        System.out.println("Result: " + result);  
    }  
}
```

Output

```
Result: true
```

Narrowing Casting/ Explicit casting

Narrowing casting must be done manually by placing the type in parentheses in front of the value:

Example#1

```
public class Main {  
    public static void main(String[] args){  
        double myDouble = 9.78;  
        int myInt =(int) myDouble; // Manual casting: double to int  
        System.out.println(myDouble); // Output 9.78  
        System.out.println(myInt); // Output 9  
    }  
}
```

Output

```
9.78  
9
```

Example#2

```
public class Main {  
    public static void main(String[] args) {  
        double numDouble = 123.45;  
        int numInt = 67;  
        int result = (int) (numDouble + numInt); // Explicit casting from double to int  
        System.out.println("Result: " + result);  
    }  
}
```

Output

```
Result: 190
```

Example#3

```
public class Main {  
    public static void main(String[] args) {  
        double numDouble = 10.5;  
        int numInt = 3;  
        boolean result = (boolean) (numDouble == numInt); // Explicit casting from double to boolean  
        System.out.println("Result: " + result);  
    }  
}
```

Output

```
Result: false
```

Exercise

- A. Why does Java allow implicit casting from **int** to **double** but requires explicit casting from **double** to **int**? Explain reason in maximum 3 lines with example code justification.
- B. Create a program that demonstrates implicit casting by:
- ❖ Assigning a **byte** value to an **int** variable
 - ❖ Assigning an **int** value to a **long** variable
 - ❖ Assigning a **float** value to a **double** variable
- Print all values before and after conversion.
- C. Create a program that demonstrates explicit casting by
- ❖ Converting a **double** to an **int**
 - ❖ Converting a **long** to a **short**
 - ❖ Converting a **float** to a **byte**
- Print all values before and after conversion
- D. Write a program that calculates the average of three integers and stores the result in an integer variable using explicit casting. Then recalculate storing the result as a double. Compare both results by finding the difference between both the results.
- E. Write a program for a shopping cart that:
- a. Takes item prices as doubles from the user input.
 - b. Calculates total as an integer (using explicit casting)
 - c. Calculates tax as a double
 - d. Shows how casting affects the final amount

Hint:

- ❖ Calculate total with explicit casting to **int**
- ❖ Calculate tax (8%) as **double**
- ❖ Calculate final amount

Expected Output:

Enter price of item 1: 19.99
Enter price of item 2: 5.50
Enter price of item 3: 12.49

Subtotal (as double): 37.98
Subtotal (cast to int): 37
Tax (8%): 2.96
Final amount: 39.96